FOR A FIELD OF CHARACTERISTIC P AN ARTIN-SCHREIER EXTENSION OF A DISCRETELY VALUED FIELD FALLS INTO ONE OF FOUR CASES LOCATED BY THE VALUATION OF THE PARAMETER

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Allowed standard prerequisites (editorial).

- Artin-Schreier theory in characteristic p. invoked at proof lines 34790-34791. Gives that the extension is Galois of order dividing p, making the four-case list exhaustive.
- Ramification theory for extensions of a discrete valuation ring, including the multiplicativity and the description of weakly unramified extensions. used at proof lines 34792-34794 and 34801-34814. Is the framework separating the cases and establishing the implications (A), (B) and (C).

Source reference. Original lemma 09I7: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 12030–12030. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original section 33172: sources/stacks-more-algebra/pinned-original-source.tex, source lines 33172–33172. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *Let A be a discrete valuation ring with fraction field K of characteristic $p > 0$. Let $\xi \in K$. Let L be an extension of K obtained by adjoining a root of $z^p - z = \xi$. Then L/K is Galois and one of the following happens*

(1) *$L = K$,*
(2) *L/K is unramified with respect to A of degree p,*
(3) *L/K is totally ramified with respect to A with ramification index p, and*

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(4) *the integral closure B of A in L is a discrete valuation ring, $A \subset B$ is weakly unramified, and $A \to B$ induces a purely inseparable residue field extension of degree p.*

Let π be a uniformizer of A. We have the following implications:

(A) *If $\xi \in A$, then we are in case (1) or (2).*
(B) *If $\xi = \pi^{-n}a$ where $n > 0$ is not divisible by p and a is a unit in A, then we are in case (3)*
(C) *If $\xi = \pi^{-n}a$ where $n > 0$ is divisible by p and the image of a in κ_A is not a pth power, then we are in case (4).*

Proof. The extension is Galois of order dividing p by the discussion in Fields, Section 09I7. It immediately follows from the discussion in Section 33172 that we are in one of the cases (1) – (4) listed in the lemma.

Case (A). Here we see that $A \to A[x]/(x^p - x - \xi)$ is a finite étale ring extension. Hence we are in cases (1) or (2).

Case (B). Write $\xi = \pi^{-n}a$ where p does not divide n. Let $B \subset L$ be the integral closure of A in L. If $C = B_{\mathfrak{m}}$ for some maximal ideal $\mathfrak{m}$, then it is clear that $p\text{ord}_C(z) = -n\text{ord}_C(\pi)$. In particular $A \subset C$ has ramification index divisible by p. It follows that it is p and that $B = C$.

Case (C). Set $k = n/p$. Then we can rewrite the equation as

$$(\pi^k z)^p - \pi^{n-k}(\pi^k z) = a$$

Since $A[y]/(y^p - \pi^{n-k}y - a)$ is a discrete valuation ring weakly unramified over A, the lemma follows. □